Instructions: Submit a Scanned handwritten Solution of the following questions before your final exam

Fundamentals of Autonomous Robots

Due Date: June 05, 2026

Course Instructor(s)

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Final Review Assignment

Sections A, B

Total Weight: 5%

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Roll No Section Student Signature

Question No 1. [Robot Vision] Provide a concise answer for each of the following questions

A. Compute the gradient direction and magnitude at the center pixel of the following image patch using Sobel operators

$$I = \begin{bmatrix} 10 & 20 & 30 \\ 15 & 25 & 40 \\ 20 & 35 & 50 \end{bmatrix}$$

B. Given the following gradient magnitude values, find the final edge map using 8-connectivity and T_H = 100 and T_L= 50

$$\begin{bmatrix} 120 & 40 & 10 \\ 90 & 70 & 20 \\ 30 & 80 & 15 \end{bmatrix}$$

C. An edge image contains two straight lines, each 200 pixels long. The Hough transform accumulator has two non-zero entries at (ρ=200, θ =0∘) and (ρ = 200, θ = 90∘). Sketch a possible edge image for this accumulator array.

D. Explain the purpose of each of the following steps performed by the canny edge detector

i) Gaussian Smoothing
ii) Gradient Direction Estimation
iii) Non-Maximum Suppression

E. Derive a single matrix representing a 3D transformation consisting of a rotation by 90^o about z-axis followed by a translation of 10 along y-axis using appropriate rotation and translation matrices

F. Derive a single matrix representing a 3D transformation consisting of a translation of 10 along y-axis followed by a rotation by 90^o about x-axis using appropriate rotation and a translation matrices

G. Give two 3 x 3 filters that can be used to compute derivatives of an image in the horizontal and vertical directions respectively

H. Give a 3 x 3 filter that can be used to smooth an image

I. Suppose we are given an over-determined system of linear equations in matrix form A.X = Y , where A is a 200 × 2 matrix, X is a 2 × 1 vector, and Y is a 200 × 1 vector. Write the closed-form expression for the least-squares solution that minimizes the mean squared error. Identify a problem that was solved using this approach.

J. Identify the connected components using **8-connectivity** in the following binary image.

$$\begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

[PCA and Face recognition]

K. A dataset consists of 4-dimensional feature vectors. After performing PCA, the following two principal eigenvectors corresponding to the largest eigenvalues and the mean of data are obtained:

Eigenvectors	Mean Vector
$v_1 = \begin{bmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 0.5 \\ -0.5 \\ 0.5 \\ -0.5 \end{bmatrix}$	$\mu = \begin{bmatrix} 2 \\ 1 \\ 1 \\ 2 \end{bmatrix}$

Use this information to reduce the dimension of the following 4D vector x to 2D. $x = \begin{bmatrix} 4 \\ 2 \\ 0 \\ 2 \end{bmatrix}$

L. Reconstruct the 4D vector from the 2D representation of previous part and compute the reconstruction error

M. List the main steps performed during training a PCA based face recognition system

N. List the main steps needed to use the PCA based face recognition system for recognizing a person.

O. State and prove the relationship between eigenvalues and eigenvectors of the matrix $A^T.A$ and $A.A^T$ where A is a matrix containing data/image vectors along its column.

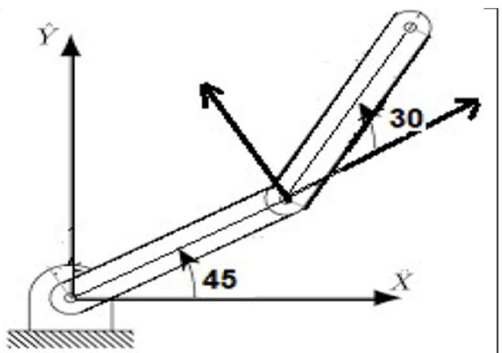
P. What is the significance of the above relationship?

[Robot Kinematics and Path Planning]

Q

Consider the following robotic arm consisting of two revolute joints. Assume that each link has length 40 cm and the C-Space configuration of the system is (45, 30) as shown in the figure on right

- **i)** Specify DH-parameters of each link?
- **ii)** Compute the transformation matrix of each link using the corresponding DH-parameters.
- **iii)** Use the transformation matrices to compute a single matrix that can be used to find the end-effectors' position for this kinematic chain.



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Question ! [Transformations]

a). Derive a transformation matrix that can be used to find position of a 2D point (x, y) after it is

i) Rotated by an angle of 90 then translated by (1, 3) and then rotated by -90.

ii) Translated by (-1 -1) then rotated by 90 and then translated by (1 1).

b). Derive a single transformation matrix for a composite transformation of a 2D-point representing

i) Rotation by an angle θ followed by Translation by $(x \ \ y)^T$.

ii) Translation by $(x \ \ y)^T$ followed by Rotation by an angle θ.

iii) Are the two transformation equivalent? Justify.

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_. For each of the following transformation matrices, determine the possible Denavit–Hartenberg (DH) parameters (a,α,d,θ) for a single-link robotic arm corresponding to the transformation.

$T = \begin{bmatrix} \cos 45^\circ & -\sin 45^\circ & 0 & 0 \\ \sin 45^\circ & \cos 45^\circ & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
$T = \begin{bmatrix} c_{30} & -s_{30} & 0 & 4c_{30} \\ s_{30} & c_{30} & 0 & 4s_{30} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$T = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 3 \end{bmatrix}$

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Two students assign different DH frames to the same manipulator and obtain different DH tables. Can both tables be correct?

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A robot operates in a 2D Euclidean configuration space with no obstacles. Its initial configuration is (3, 4), and the goal configuration is (6, 7).

Using

- **gradient descent** as the path planner with step size $\alpha = 0.1$ and gain $k = 1$
- **quadratic** attractive potential

Compute the **first three configurations** generated by the path planner starting from the initial position. Show all steps including gradient calculation and update rule.

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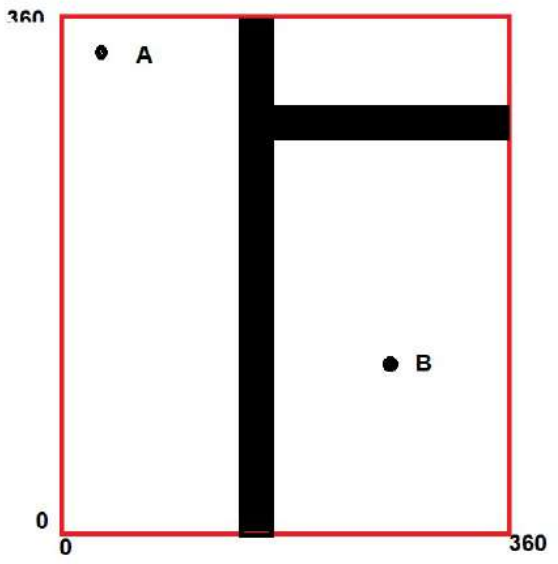
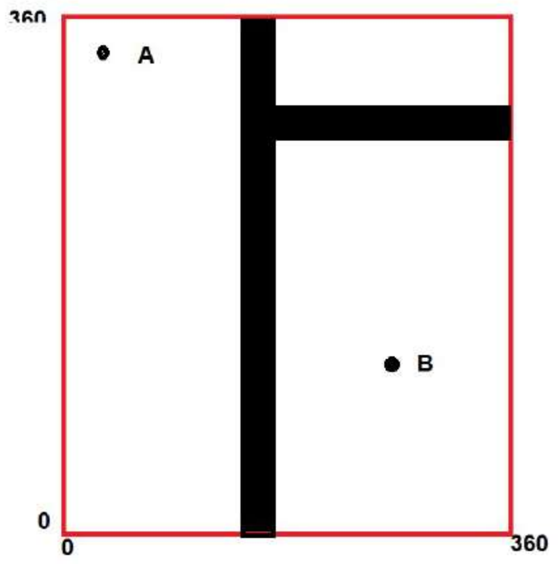
Q N [Configuring Space]

Consider the following very similar configuration spaces for two robotic arms one having two revolute joints while the other with two prismatic joints.

C-Space of arm with Revolute Joints	C-Space of arm with prismatic Joints
	

Part a) Find a path from Points A to B for the robot with Revolute joints and draw the starting and ending configurations of the robotic arm assuming that both links have same length

Part b) Explain why it is not possible to reach from A to B if the C-Space of robot with prismatic joints is used

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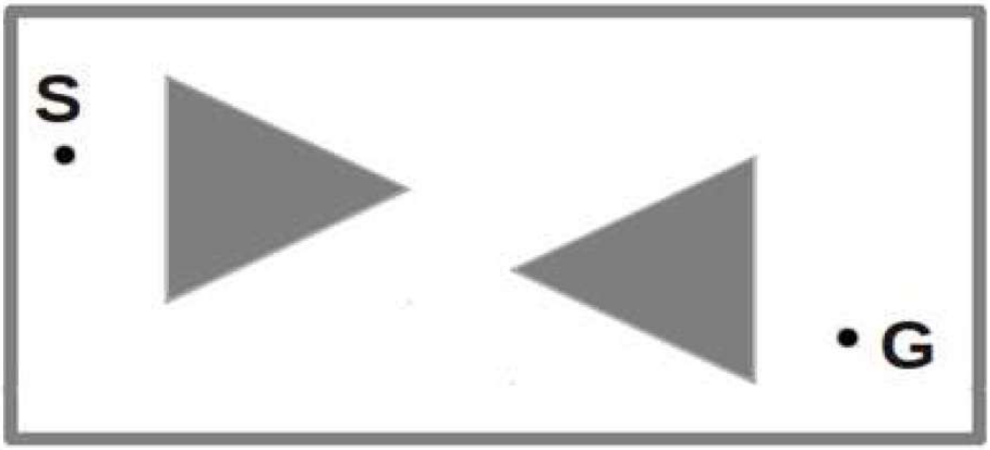
Q **Part a)** A 2D configuration space corresponding to working space of a Robot is shown below. i) **On the figure, draw all supporting and separating** edges ii) Use the resulting graph to find the shortest path between S and G using Dijkstra Algorithm. Use length of each edge in millimeters as the weight of that edge**. Show the working of Dijkstra on the answer script and mark the path found on this figure**



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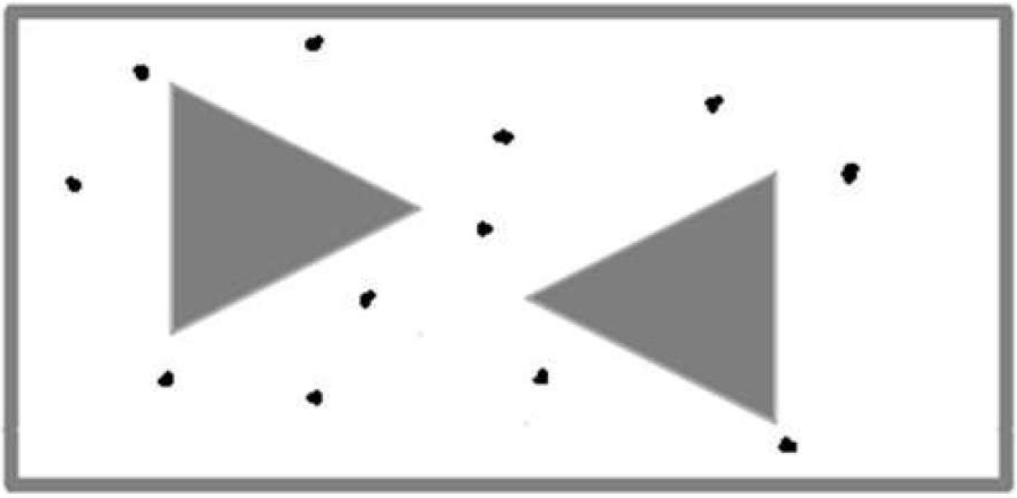
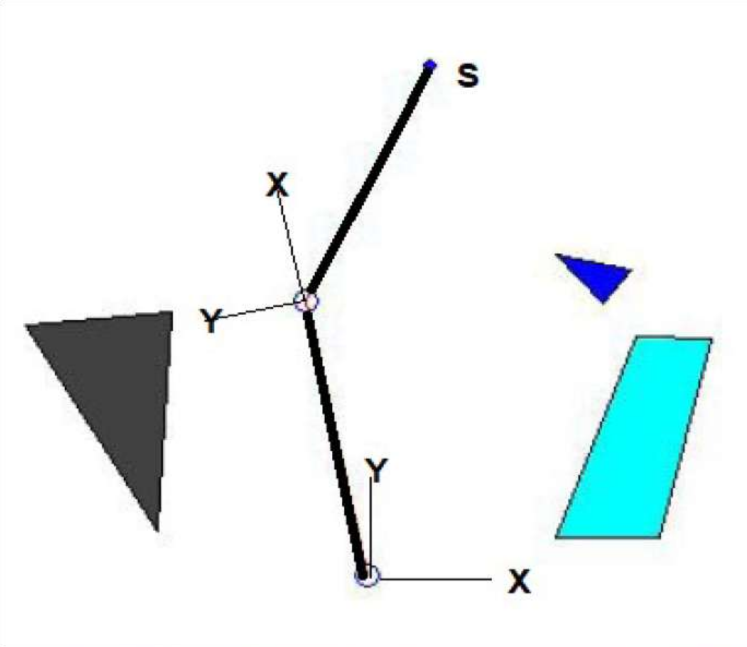
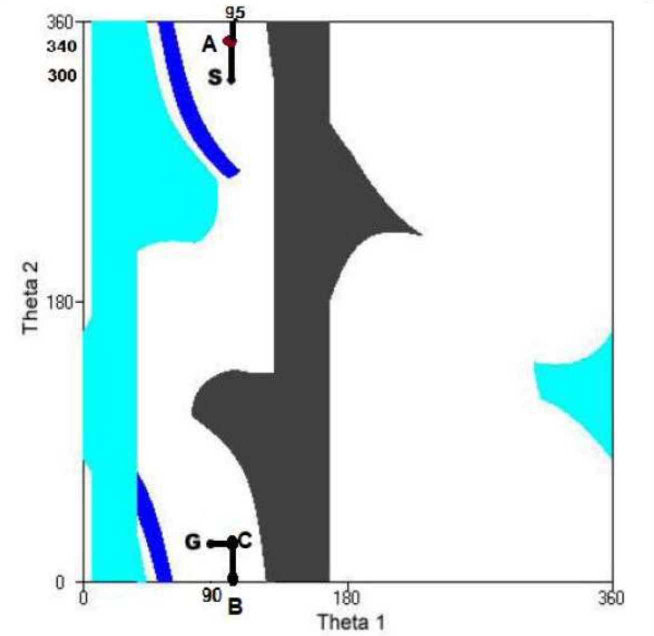
Part b) Use K = 3 (i.e. three nearest neighbors) to build and show the probabilistic roadmap (PRM) using the 12 randomly selected points shown on the configuration given below.



Y

Q – The working space of a 2-link kinematic chain Robot with two revolute joints and the corresponding configuration space is shown below. Theta 1 is the angle of first joint w.r.t. frame 0 and Theta 2 is angle of second joint w.r.t. frame 1. S is the initial configuration of the Robot. A path from S to G is shown with intermediate states A, B and C.

i) On the figure, draw the working space state (without the reference frames) corresponding to the configuration space points A, B, C and G.

Working Space	**Configuration Space**
	

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ii) Another path from S to G in the configuration space is shown below. Draw the configuration of the Robot in the working space corresponding to the points A, B, C, D, E, and G marked on this path.

Working Space	**Configuration Space**